



Lime – Properties, Types & Uses

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Introduction to Lime

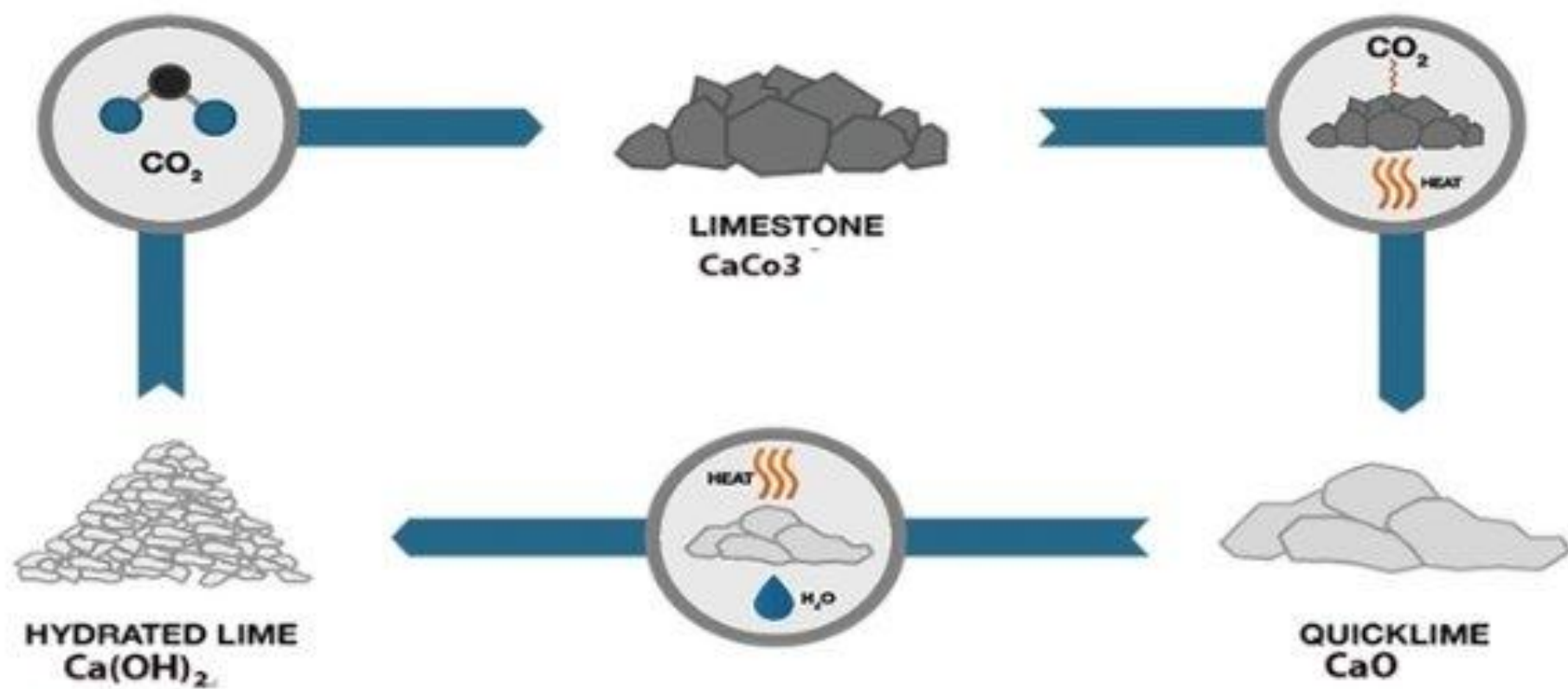
- The word **lime** refers to products derived from burnt (calcined) limestone, such as quicklime and hydrated lime.
- **Limestone** is a naturally occurring and abundant sedimentary rock consisting of high levels of **calcium** and/or **magnesium carbonate**, and/or **dolomite** (calcium and magnesium carbonate), along with small amounts of other minerals.
- It is extracted from quarries and underground mines all over the world.
- Lime and limestone products are among the oldest materials used by humans for a very diverse range of applications.
- Today these products serve as an essential building block in every industrial process.



Introduction to Lime

- The ancient Romans used lime in building and road construction, which continues to the present day.
- In earlier days, lime was made by heating limestone (calcium carbonate) to high temperatures.
- This process, known as calcining, results in quicklime or calcium oxide.
- Hydrated lime (calcium hydroxide) is produced by reacting quicklime with sufficient water to form a dry white powder.
- While lime is one of the earliest industrial commodities known to man, its production and uses have grown with the times, and it continues to be one of the earliest building blocks of modern industry.
- Although cement has replaced lime to a great extent, yet in view of the rising cost, it is quite likely that lime will come into prominence again.

Quick Lime vs. Hydrated Lime: Key Differences Explained

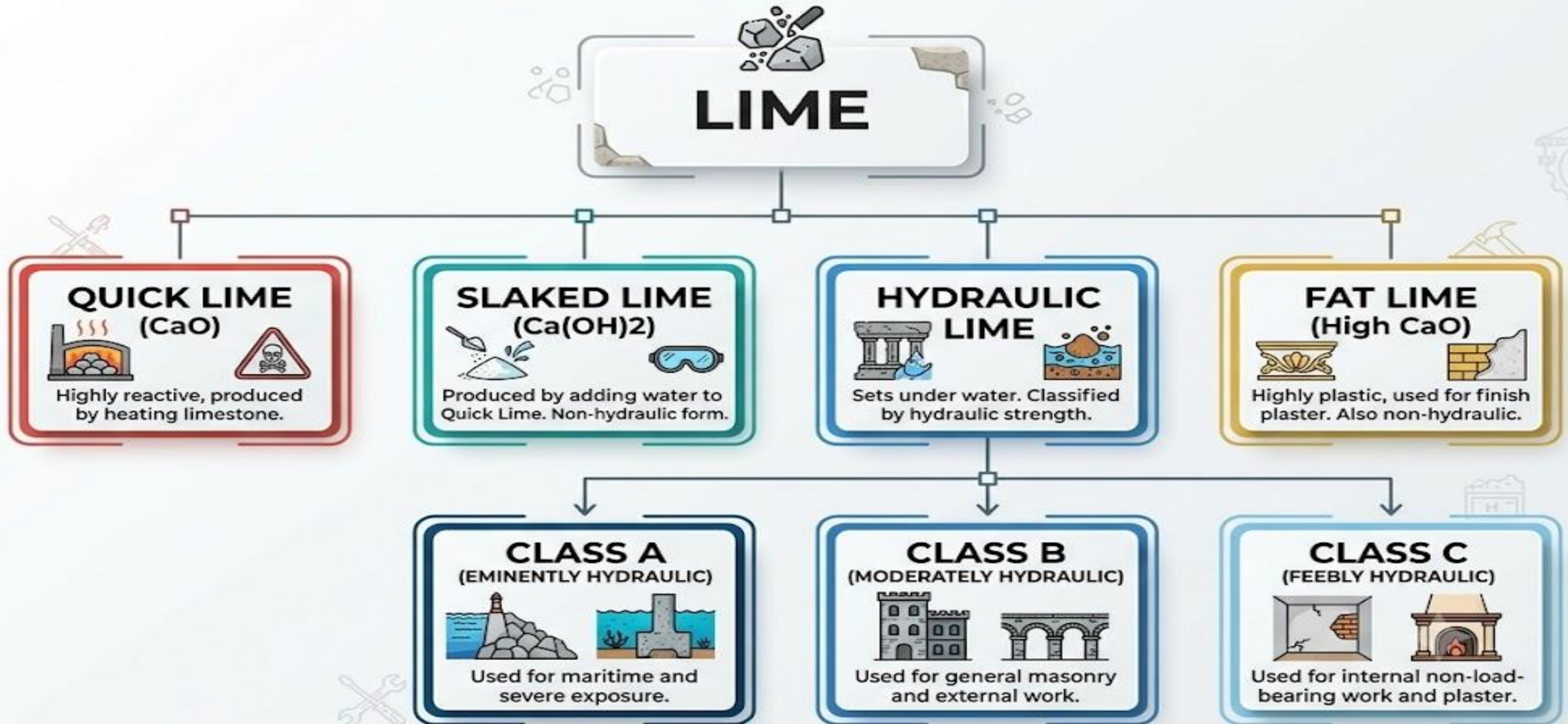




Properties of lime

- ❖ It should possess good plasticity.
- ❖ It should be flexible and easily workable.
- ❖ When used in mortar, it should provide greater strength to the masonry.
- ❖ It should solidify in less time and become hard.
- ❖ It should comprise of excellent binding properties which adhere to brick or stone masonry units perfectly.
- ❖ It possesses high durable properties as it is less shrinkable when used in mortar.
- ❖ It should be highly resistant to moisture and can be used for pointing works.

Types and Classification of Types and Construction of Lime





Types of Lime

- ❖ **Quick lime:** It's obtained by calcination (i.e. heating to redness) of comparatively pure limestone. It's amorphous in nature, highly caustic and possesses great affinity to moisture.
- ❖ **Slaked lime (or hydrated lime):** It's obtained by slaking (i.e., combination of quick lime with water) of quick lime. It has got the tendency of absorbing carbonic acid from the atmosphere in presence of water and it's an ordinary pure lime in white powder form.
- ❖ **Fat lime (or high calcium lime or pure lime or rich lime or white lime):** It's obtained from the pure limestone, shell and coral. It absorbs carbon dioxide when it is left in air, and gets transferred into calcium carbonate.



Types of Lime

- ❖ When compared with quick lime, the volume of fat lime gets increased to about 2-2.5 times.
- ❖ It's used for various purposes as whitewashing, plastering of walls, as a lime mortar with sand for pointing in masonry work, as a lime mortar with surkhi for thick masonry walls, foundations, etc.,
- ❖ **Hydraulic lime (or water lime):** The composition of hydraulic lime is different from quick lime; it contains a definite quantity of clay which gives its hydraulic property.
- ❖ Hydraulic lime contains 70-80% of CaOH and 15-30% of clay.
- ❖ It has the capacity to set and harden even under water, where the quick lime can't be used and hence its also known as water lime.
- ❖ It's obtained by burning the limestone containing rich clay or adding clay materials to the limestone while burning.



Types of Lime

- ❖ Depending upon the hydraulicity, its divided into feebly hydraulic, moderately hydraulic and non-hydraulic lime.
- ❖ **CLASS A Lime (Eminently hydraulic lime):** It contains about 25% clay content and sets readily under water within a day or so. It slakes with difficulty. The mortar and lime concrete prepared from this lime is very useful for construction under water or in damp places.
- ❖ **CLASS B Lime (Semi hydraulic or moderately hydraulic lime):** It contains about 15% clay and sets under water at a slower rate within a week or so. The mortar and concrete prepared from this lime is strong and used for superior type of masonry work.
- ❖ **CLASS C Lime (Non-hydraulic lime):** It contains about 7-5% clay and is prepared from pure limestone. This slakes vigorously within a few minutes but doesn't set under water. It is used for whitewashing and color washing.



Types of Lime

Parameter	Fat Lime (Non-Hydraulic Lime)	Hydraulic Lime (Water Lime)
Composition	Pure limestone containing about 95% CaO with less than 5% clay impurities.	Contains 70 - 80% Ca(OH) and 15 – 30% clay (silica/alumina).
Hydraulicity	Does not possess hydraulic properties (cannot set underwater or without air).	Possesses hydraulic properties (sets and hardens underwater/in damp places).
Setting Action	Sets very slowly by absorbing CO from the atmosphere (carbonation).	Sets faster through a chemical reaction with water (hydrolysis).
Slaking Action	Slakes vigorously with a high release of heat and noise.	Slakes slowly with minimal heat and no cracking sound.
Volume Change	Volume expands significantly (2 to 2.5 times during slaking).	Volume increases very little compared to fat lime.
Colour	Perfectly white.	Off-white / brownish / dull shade due to clay content.
Strength & Durability	Low mechanical strength; soft and flexible.	Stronger, more durable, and similar to low-grade cement.
Primary Uses	Whitewashing, plastering, decorative finishes, and thin masonry work.	Structural masonry, underwater construction, foundations, and damp environments.



Uses of Lime

- ❖ Lime is one of the most versatile and historically significant binding materials in construction. Key applications across structural, finishing, and civil engineering include:
- ❖ **Brick & Stone Laying:** Lime mixed with sand forms lime mortar, providing high workability, strong adhesion, and flexible joints that prevent masonry units from cracking under building movement.
- ❖ **Lime-Surkhi Mortar:** Blended with surkhi (powdered burnt brick), it creates a hydraulic binder suitable for thick masonry walls and footings.
- ❖ **Plastering & Pointing:** Lime plasters stick easily to walls and provide a breathable finish that regulates internal humidity, preventing moisture entrapment.
- ❖ **Whitewashing & Tinting:** Fat lime or Class C lime is mixed with water to create limewash, acting as a decorative, anti-bacterial, and breathable coating for interior and exterior walls.



Uses of Lime

- ❖ **Soil Stabilization:** Quicklime or hydrated lime is mixed into clayey or expansive soils to dry them out, reduce plasticity, and significantly increase load-bearing capacity for road bases and building site prep.
- ❖ **Underwater Construction:** Eminently hydraulic lime (Class A) sets underwater without air, making it ideal for marine structures, dams, and foundations in damp places.
- ❖ **Sand-Lime Bricks:** Lime reacts with silica sand under steam pressure to manufacture uniform, durable calcium-silicate bricks.
- ❖ **Autoclaved Aerated Concrete (AAC):** Used as a primary binder in AAC blocks to provide lightweight, highly insulated structural units.
- ❖ **Cement Production:** Calcined quicklime serves as a fundamental raw material in the manufacturing process of Portland cement.

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